
A3 Constructa

Maintenance guide — how it is built and how to change it

For the team maintaining the app. Covers how the code is laid out, what each workspace contains, the rules that keep changes reproducible, and the traps that have already cost us time.

Acube Innovations Pvt Ltd — internal

1. How the app is put together

Everything lives in one Frappe app, **a3_constructa**, under a single module also called **A3 Constructa**. One module was chosen deliberately so that **Module = A3 Constructa** in the DocType list shows everything the app owns — the same way A3 Dumas, Optics and Retail are built.

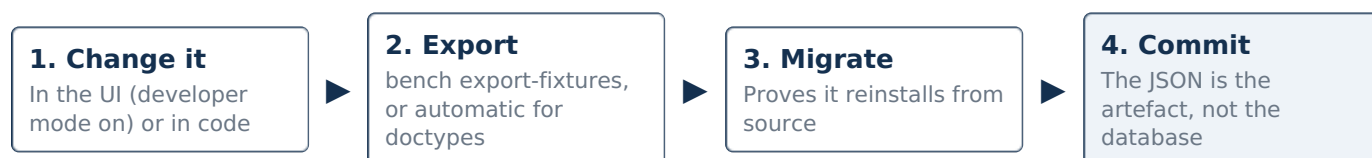
```
a3_constructa/
├── a3_constructa/      the single module
│   ├── doctype/        24 doctypes (19 parent + 5 child)
│   ├── report/         68 reports (script + report builder)
│   ├── number_card/    7 finance KPI cards
│   ├── dashboard_chart/ 5 charts
│   ├── a3_constructa_dashboard/ the Construction Finance dashboard
│   └── workspace/      10 workspaces (9 + the parent index)
├── overrides/          hooks into core doctypes
│   ├── stock_entry.py  cost-code accounting, MIN/MRN series
│   ├── asset.py        per-category asset numbering
│   └── serial_no.py    fills item_group on serials
├── setup/install_defaults.py records built in code, not fixtures
├── demo/               the demo project, by stage
├── fixtures/           exported master data (see rules overleaf)
├── docs/               this guide and the client guide
└── hooks.py            fixtures list, doc_events, required apps
```

What goes where

Kind of change	Lives in	Why
DocType, report, workspace	module folder, as JSON	Frappe exports these automatically while developer mode is on.
Custom field, property setter	fixtures/*.json	Exported by bench export-fixtures , filtered by module.
Master records	fixtures/*.json	Item Group, Stock Entry Type, Warehouse Type, Document Type, Workflow.
Records needing a company	setup/install_defaults.py	Asset Category and the Retention account carry company-specific accounts, so a fixture would hard-code this site's chart of accounts.
Behaviour on a core doctype	overrides/ + hooks.py	Never edit erpnext or frappe. Hook it.

The change pipeline

Every change follows the same path from your edit to version control.



Order matters. Export before you migrate, never after. Migrate re-imports the fixtures on top of the database, so a stale export silently reverts whatever you just changed. This has bitten us twice — see section 5.

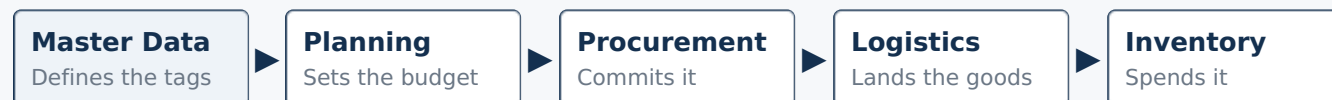
2. The nine workspaces

All nine sit under the **A3 Constructa** parent in the sidebar. They map one-to-one onto the build sheet's 350 rows.

Workspace	Rows	What it owns	Worth knowing
Master Data	77	Cost Head, WBS, Cost Code, Port, Service Route, Freight Rate Contract, Document Type, Mandatory Document Rule, Approval Matrix	Cost Head and WBS are nested-set trees. 15 report-builder reports back its filtered card links.
Planning & Budgeting	21	BOQ (+ items), WBS Allocation, Procurement Plan	BOQ is submittable and gated by the BOQ Approval workflow. Budget figures for every other report start here.
Procurement	31	Work Certificate (+ items)	Mostly custom fields on Material Request Item, Supplier Quotation and Purchase Order Item. Retention is calculated on the certificate.
Procurement Classification	14	Item Groups, Asset Categories	Almost entirely master records. Converts ERPNext's stock "Services" group into a parent group.
Delivery & Logistics	30	Shipment Tracking, Shipment Milestone, Shipment Document	One doctype assembled from seven rows. Blocking document rule fires on submit.
Inventory Movement	29	No doctypes — Stock Entry Types, custom fields, 10 reports	Two hooks on Stock Entry: cost-code accounting and the MIN/MRN series.
Asset & Equipment	43	Asset Spare Part, Tool Issue (+ items)	Tool Issue posts a Stock Entry on submit and cancels it on cancel. Assets number per category via a hook.
HR & Time	42	No doctypes — 8 custom fields, 3 reports	Everything is hrms. The fields give labour a project dimension; Attendance had none at all.
Finance & Accounting	63	No doctypes — 4 script reports, 7 number cards, 6 charts, 1 dashboard	Cost Code Wise Costing is the report everything else reads from. Do not rename its columns.

How data flows between them

THE SPINE — every stage tags project, cost head, WBS and cost code



FEEDING THE SAME COST STRUCTURE



The one dependency to remember: **Cost Code Wise Costing** reads budget from WBS Allocation, commitments from Purchase Order Item and actuals from both GL Entry and the stock ledger. **Cost to Complete** imports its three helper functions directly. Change one and check the other.

3. The rules that keep it reproducible

The app has to install cleanly on a fresh site with erpnext and hrms. These six rules are what make that true.

1. The exported JSON is the source of truth, not the database

For fixture doctypes, `bench migrate` imports the JSON *over* the database. If you change a record in the UI and migrate before exporting, your change is gone. Export first.

2. Never list a Workspace as a fixture

A Workspace with a module is already exported to the module folder and synced by migrate. Listing it in `fixtures` as well gives it two sources of truth, and the fixture wins because it imports last.

3. Filter every master-data fixture by an explicit allow-list

Item Group, Stock Entry Type, Warehouse Type and the workflow doctypes have no `module` field. An unfiltered entry would export every stock ERPNext Item Group into our app and reinstall them everywhere. Each list in `hooks.py` is the app's inventory of the master data it owns — add a name whenever you create a record.

4. Anything company-specific goes in code, not fixtures

Account names carry the company abbreviation and parents differ by chart of accounts. Asset Category and the Retention Payable account resolve theirs at install time in `setup/install_defaults.py`, which runs on `after_install` and again on `after_migrate` and is written to be idempotent.

5. Do not edit erpnext or frappe — hook them

Three hooks exist today, all in `overrides/` and registered in `doc_events`. Follow the same pattern.

6. Freeze report column names that a card or chart binds to

A Number Card or Dashboard Chart of type Report names a column by fieldname. Rename the column and the card renders empty **with no error**. The two bound today are `percent_consumed` on Cost Code Wise Costing and `committed_amount` on Budget vs Commitment.

Verify before you commit

```
bench --site <site> migrate      # must be clean
bench export-fixtures --app a3_constructa
bench --site <site> migrate      # again: changes must survive
git status                      # fixtures should be unchanged the 2nd time
```

4. How to add or change things

Add a custom field to a core doctype

Create it with `create_custom_fields` and set `module = "A3 Constructa"` — the fixture filter keys off that, and a field without it will not be exported.

```
from frappe.custom.doctype.custom_field.custom_field import create_custom_fields

create_custom_fields({
    "Purchase Order Item": [
        {
            "fieldname": "site_zone",
            "label": "Site Zone",
            "fieldtype": "Data",
            "insert_after": "cost_code",
            "module": "A3 Constructa",
        },
    ],
}, ignore_validate=True)

# then: bench export-fixtures --app a3_constructa && bench migrate
```

Add a DocType

Create it in the UI with developer mode on, module **A3 Constructa**. Frappe writes the JSON, a controller stub and a test file into the module folder as you save. Commit all of them. For a tree, set `is_tree` and `nsm_parent_field` and add `lft`, `rgt`, `old_parent`; the controller must extend `NestedSet`.

Add a report

Script Report for anything that aggregates or joins; Report Builder for a filtered list view. Set `is_standard = Yes` and the module, and Frappe exports it. A Report Builder report is the only way to give a workspace card link a filter — Workspace Link has no filter field.

Add a link to a workspace card

Edit the workspace in the UI, or the JSON in `a3_constructa/workspace/<name>/. A link can point at a DocType, a Page or a Report — not a Dashboard. For a dashboard, use a Workspace Shortcut instead.`

Add a master record that must exist everywhere

Create the record, add its name to the matching allow-list in `hooks.py`, then export fixtures. If it needs a company or an account, put it in `install_defaults.py` instead.

Add or change a workflow

Workflow, Workflow State and Workflow Action Master are fixtures with allow-lists. Only add names we introduce — Approved, Rejected, Approve and Reject ship with Frappe and must not be re-exported. Drive a workflow from code with `apply_workflow(doc, "Action")`; setting the state field directly is refused, which is the point of it.

Hook behaviour onto a core doctype

Add a function in `overrides/` and register it in `doc_events`. Pick the event carefully — see the traps overleaf.

Add a new workspace

Set `parent_page = "A3 Constructa"` and the next `sequence_id`, module **A3 Constructa**, and `public = 1`. Cards are **Card Break** rows in the `links` table followed by their **Link** rows; the `content` JSON lists a card block per card.

5. Traps we have already hit

Each of these shipped silently — no error, wrong behaviour. They are listed because the same shape of mistake will recur.

A stale fixture reverted a database change on migrate

- Symptom** Workspaces un-nested themselves; later, 17 custom fields pointed at deleted modules and fell out of version control entirely.
- Cause** Fixture import runs *after* the module sync on every migrate. The exported JSON was older than the database.
- Fix** Export before migrating. Workspace removed from the fixtures list altogether.

A validate hook never won, because ERPNext validated first

- Symptom** Material issues kept posting to Stock Adjustment instead of the cost code's account.
- Cause** `doc_events` validate runs after the controller's own validate, which had already filled the field. Our "only if empty" guard meant the cost code never applied.
- Fix** Overwrite deliberately when a cost code is set, and say so in the code.

A naming series set on validate is already too late

- Symptom** Issue notes numbered MAT-STE- even though MIN- was in the options list.
- Cause** The name is built in `set_new_name`, before validate runs.
- Fix** Use the `before_naming` event, which frappe calls from inside `set_new_name`. Same fix for per-category asset numbering.

A report bound to a column that does not exist

- Symptom** Project Profitability chart rendered empty, no error.
- Cause** Bound to `gross_profit_percent`; the report calls it `profit`.
- Fix** Check the source report's fieldnames before binding a card or chart to them.

A field ERPNext never populates

- Symptom** The small-tools register was empty on every site, not just ours.
- Cause** `Serial No.item_group` exists but is never filled — serials come from stock movements carrying only an item code.
- Fix** A `before_insert` hook copies it from the item, plus a backfill for existing serials.

The pattern: creating a record proves nothing. Every one of these passed "does it exist". They only showed up when a real transaction was posted and the numbers were read back. Test with data.

6. Commands and where things live

Everyday

```
docker compose up -d          # start the stack
docker compose logs -f backend # watch it
docker compose exec backend bash # shell in the bench

bench --site constructa.localhost migrate
bench --site constructa.localhost clear-cache
bench --site constructa.localhost console
bench export-fixtures --app a3_constructa
```

The demo

```
bench --site constructa.localhost execute \
  a3_constructa.demo.construction_demo.run

bench --site constructa.localhost execute \
  a3_constructa.demo.construction_demo.run --kwargs '{"stage':'finance'}"

bench --site constructa.localhost execute \
  a3_constructa.demo.construction_demo.teardown
```

run is idempotent and takes an optional stage: masters, planning, procurement, logistics, inventory, plant, people, finance, extras. **teardown** reports what it could not remove — warehouses and serial numbers are pinned by stock ledger entries, and the clean answer there is to drop the site.

The documents

```
bench --site constructa.localhost execute a3_constructa.demo.build_docs.build_pdf
bench --site constructa.localhost execute a3_constructa.demo.build_docs.build_guide
```

The HTML in **docs/** is the source; the PDF is the output. Do not edit the PDF.

7. Environment and known gaps

Ports on this machine

Service	Port	Note
A3 Constructa	8092 / 8003 / 3309	Browser, dev server, MariaDB
a3-optics	80 / 8000 / 3306	Separate bench
a3-retail	8090 / 8001 / 3307	Separate bench
a3-loan	8091 / 8002 / 3308	Separate bench
erpnext15	8010	Separate bench

Known gaps

Item	Why it is open
Subcontract PO report	Needs ERPNext subcontracting configured with a finished-good item and Subcontracting BOM. Work Certificate covers the payment path instead.
Transfer Despatch Notes	Needs an internal customer defined.
Loans (HR row 24)	Loan Management moved to the separate lending app, not installed.
13 label-only links	Master Data cards where the sheet gives no filter basis. They work, but siblings open the same list.
GL Entry wbs / cost_code	Fields exist; nothing populates them yet. Budget reports read actuals from the stock ledger in the meantime.

Branch discipline: work happens on [dev-muzzammil](#). If anyone edits the app on [main](#), the next merge stops being free.